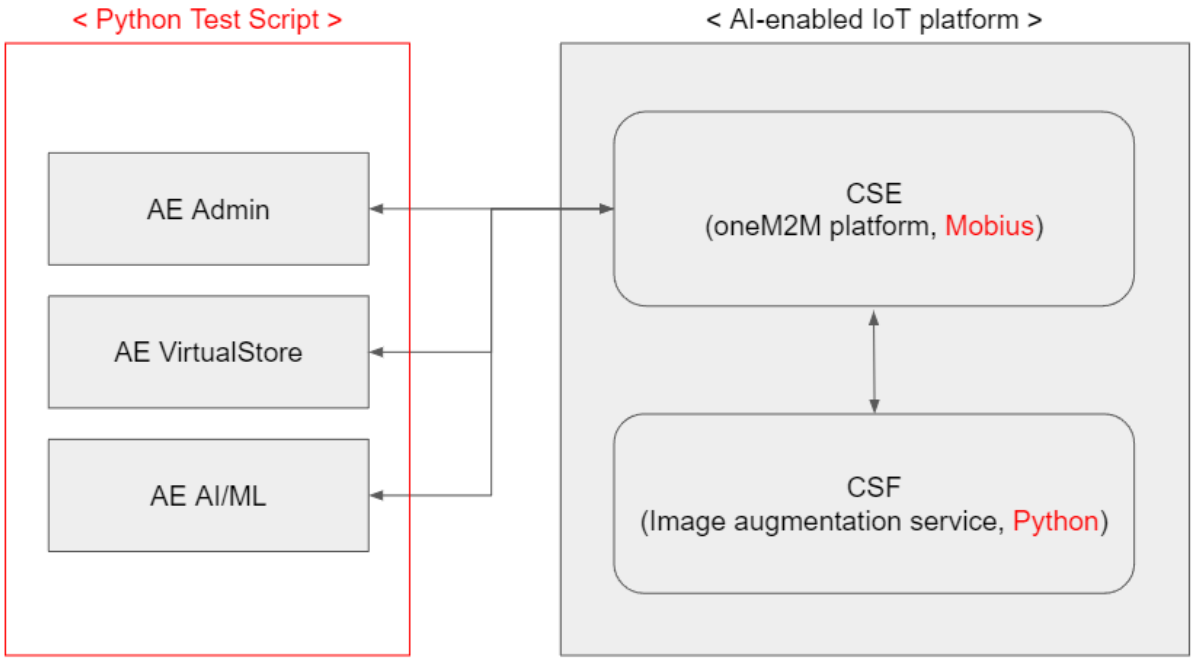
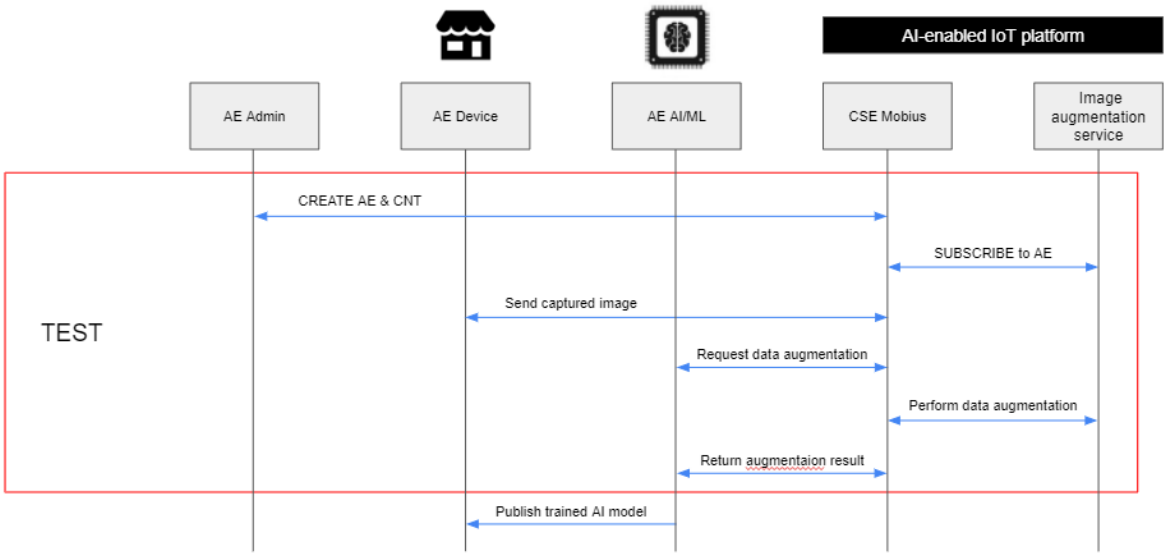
Data Augmentation test

Test Script 실행 순서
Mobius 실행
virtualStore.py
Create_cnt.py
Create_water_img1.py
CSF.py
Create_da.py
Training.py



Test Script 실행 순서



1. Mobius의 경로 이동합니다.
2. `node mobius` 명령어를 통해 Mobius를 실행합니다.
3. virtualStore.py를 실행합니다.
4. Create_cnt.py를 실행합니다.
5. Create_water_img1.py를 실행합니다.
6. CSF.py를 실행합니다.
7. Create_da.py를 실행합니다.
8. training.py를 실행합니다.

Mobius 실행

```
fabu1ous@ubuntu: ~/Desktop/work/mobius-dataAug
fabu1ous@ubuntu:~/Desktop/work/mobius-dataAug$ node mobius
Production Mode
Garbage collection is not exposed
sgn_mqtt_client is connected
CPU Count: 2
select_ri_lookup /Mobius: 1.489ms
update_cb_poa_csi /Mobius: 0.672ms
""
Production Mode
Garbage collection is not exposed
sgn_mqtt_client is connected
mobius server (192.168.0.5) running at 7579 port
select_ri_lookup /Mobius: 1.589ms
update_cb_poa_csi /Mobius: 0.508ms
""
Production Mode
Garbage collection is not exposed
sgn_mqtt_client is connected
mobius server (192.168.0.5) running at 7579 port
select_ri_lookup /Mobius: 1.683ms
update_cb_poa_csi /Mobius: 0.680ms
""
pxyws server (192.168.0.5) running at 7577 port
```

virtualStore.py

virtualStore(ae) 리소스를 CREATE하고 child 리소스에 SUBSCRIBE하는 코드입니다.

virtualStore와 sub_da 리소스에 대한 CREATE 성공 response가 화면에 출력됩니다.

```
fabu1ous@ubuntu:~/Desktop/work/mobius-dataAug/test_scripts/AI$ python3 virtualSt
ore.py
{"m2m:ae":{"rn":"virtualStore","ty":2,"pi":"5-20211014162943715688","ri":"Svirtu
alStore","ct":"20220504T051001","lt":"20220504T051001","et":"20240504T051001","a
pi":"0.2.481.2.0001.001.000111","rr":true,"lbl":["virtualStore"],"aei":"Svirtual
Store"}}
{"m2m:sub":{"rn":"sub_da","ty":23,"pi":"SvirtualStore","ri":"23-20220504051001305877","ct":"20220504T051001","lt":"20220504T051001","et":"20240504T051001","nu":
["mqtt://localhost:7579/virtualStore/da?ct=json"],"enc":{"net":[3]},"exc":100,"n
ct":2,"cr":"SvirtualStore"}}
```

Create_cnt.py

테스트에 필요한 cnt리소스들을 생성해주는 코드입니다.

`/virtualStore/classifier/target` 리소스를 생성합니다.

classifier와 target리소스에 대한 CREATE 성공 response가 화면에 출력됩니다.

```
fabu1ous@ubuntu:~/Desktop/work/mobius-dataAug/test_scripts/AI$ python3 Create_cn
t.py
{"m2m:cnt":{"rn":"classifier","ty":3,"pi":"SvirtualStore","ri":"3-20220504051714
720575","ct":"20220504T051714","lt":"20220504T051714","st":0,"et":"20240504T0517
14","mni":3153600000,"lbl":["classifier"],"mbs":999999,"mia":31536000,"cr":"Svir
tualStore","cni":0,"cbs":0}}
{"m2m:cnt":{"rn":"target","ty":3,"pi":"3-20220504051714720575","ri":"3-202205040
51714733867","ct":"20220504T051714","lt":"20220504T051714","st":0,"et":"20240504
T051714","mni":3153600000,"lbl":["target"],"mbs":999999,"mia":31536000,"cr":"Svi
rtualStore","cni":0,"cbs":0}}
```

Create_water_img1.py

`/virtualStore/classifier/target/water_img1` 리소스를 생성하는 코드입니다.

water_img1(cin)은 Augment의 대상이 될 source resource입니다.

```
fabu1ous@ubuntu:~/Desktop/work/mobius-dataAug/test_scripts/AI$ python3 Create_wa
ter_img1.py
{'m2m:cin': {'rn': 'water_img1', 'ty': 4, 'pi': '3-20220504051714733867', 'ri':
'4-20220504051943970782', 'ct': '20220504T051943', 'lt': '20220504T051943', 'st'
: 1, 'et': '20240504T051943', 'cs': 62544, 'con': '/9j/4AAQSkZJRgABAQEBLAEsAAD/2
wBDAAICAgICAQICAgIDAgIDAwYEAwMDAwcFBQQGCAcJCAgHCAgJCg0LCQoMCggICw8LDA0ODg8OCQsQE
RAOEQ0ODg7/2wBDAQIDAwMDAwcEBAcOCQgJDg4ODg4ODg4ODg4ODg4ODg4ODg4ODg4ODg4ODg4ODg4OD
```

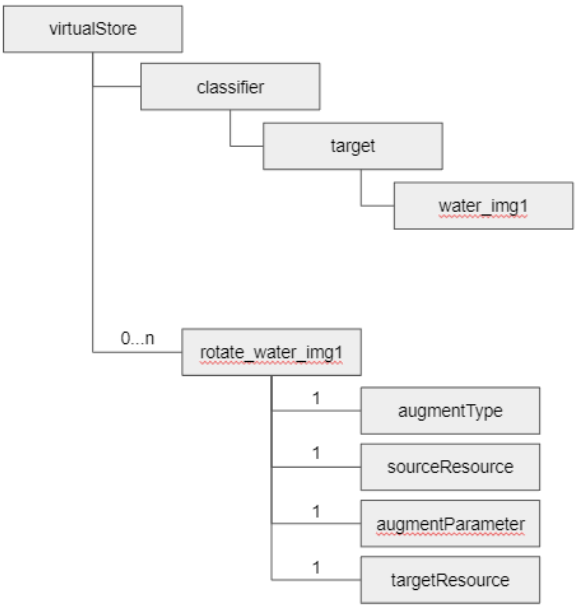
CSF.py

Image Augmentaion Service에 해당하는 코드입니다.

화면에 listening... 이 출력되면 성공입니다.

```
fabu1ous@ubuntu:~/Desktop/work/mobius-dataAug/test_scripts/AI$ python3 CSF.py
listening...
Connected to MQTT brocker
```

Create_da.py



1 Request data augmentation

```
POST /Mobius/AI HTTP/1.1
Host: xxx.xxx.xxx.xxx:7579
Accept: application/json
X-M2M-RI: 12345
X-M2M-Origin: dataAugmentation
Content-Type: application/vnd.onem2m-res+json; ty=50
{
        "m2m:da": {
                        "rn": "rotate_water_img1",
                        "augty": "rotate",
                "srsrc":
        "/virtualStore/classifier/target/water_img1",
                "augprm": {
                        "max_left_rotation": "20",
                        "max_right_rotation": "20",
                        "sample": "20",
                        "label": "water"
                },
                "trgrsrc": "/virtualStore/classifier/target"
                }

}
```

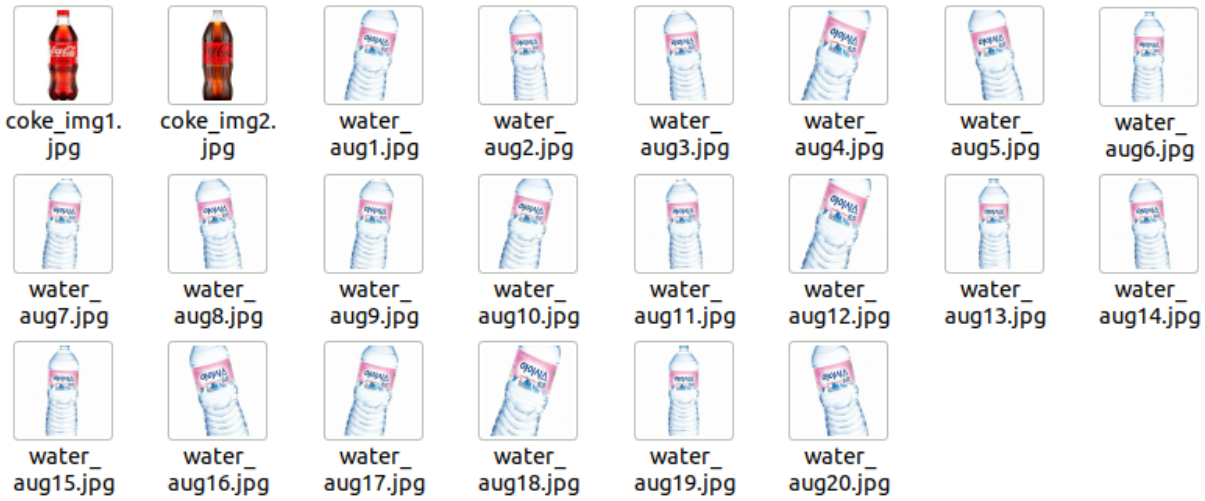
rotate_water_img1(da)리소스를 생성하는 코드입니다.

```
fabu1ous@ubuntu:~/Desktop/work/mobius-dataAug/test_scripts/AI$ python3 CSF.py
listening...
Connected to MQTT brocker
message topic= /oneM2M/req/Mobius2/virtualStore/da/json
/virtualStore/classifier/target/water_img1
Initialised with 1 image(s) found.
Processing <PIL.Image.Image image mode=RGB size=500x500 at 0x7F4AABD9B5B0>: 100%
water

fabu1ous@ubuntu:~/Desktop/work/mobius-dataAug/test_scripts/AI$ python3 Create_da
.py
{"m2m:da":{"rn":"rotate_water_img1","ty":50,"pi":"SvirtualStore","ri":"50-202205
04053401422545","ct":"20220504T053401","lt":"20220504T053401","st":0,"et":"20240
504T053401","augty":"rotate","srsrc":"/virtualStore/classifier/target/water_img1
","augprm":{"max_left_rotation":"20","max_right_rotation":"20","sample":"20","la
bel":"water"},"trgrsrc":"/virtualStore/classifier/target","cr":"SvirtualStore"}}
fabu1ous@ubuntu:~/Desktop/work/mobius-dataAug/test_scripts/AI$
```

rotate_water_img1에 대한 CREATE 성공 repsonse를 화면에 출력합니다.

rotate_water_img1가 CREATE 되면 Image Augmentaion Service(CSF)에서 Augment를 수행합니다.

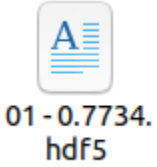


/mobius/test_script/train 경로에서 augment된 이미지들을 확인할 수 있습니다.

Training.py

water와 coke 이미지를 구분하는 모델을 학습시키는 코드입니다.

/mobius/test_script/model 경로에 모델을 생성합니다.



```
Epoch 1: val_loss improved from inf to 0.86559, saving model to ./model/ 01 - 0.
8656.hdf5
32/32 [==============================] - 3s 41ms/step - loss: 1.2525 - accuracy:
 0.3333 - val_loss: 0.8656 - val_accuracy: 0.5000
```

Data Augment 전 training.py의 val_loss와 val_accuracy입니다.

```
Epoch 1: val_loss improved from inf to 0.77344, saving model to ./model/ 01 - 0.
7734.hdf5
32/32 [==============================] - 8s 178ms/step - loss: 0.4340 - accuracy
: 0.8462 - val_loss: 0.7734 - val_accuracy: 0.8889
```

Data Augment 후 training.py의 val_loss와 val_accuracy입니다.